Notes - Implementing an Oscillation Force due to ISS Vibration

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1 Non-dimensionalize one-way coupled particle equation of motion

Author: Bernhard (30. March 2020), Fabian (12. May 2020)

For the present simulations, we solve the unsteady Navier-Stokes equations for an incompressible Newtonian fluid, given by

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) = -\frac{1}{\rho_f}\nabla p + \nu_f \nabla^2 \mathbf{u} + \mathbf{f}_{IBM} + \mathbf{f}_{osc} \quad , \tag{1}$$

and the continuity equation, given by

$$\nabla \cdot \mathbf{u} = 0 \quad , \tag{2}$$

on a uniform rectangular grid with grid cell size $\Delta x = \Delta y = \Delta z = h$. Here, $\mathbf{u} = (u, v, w)^T$ designates the fluid velocity vector in Cartesian components, p the pressure, ν_f the kinematic viscosity, t the time, $\mathbf{f}_{IBM}$ an artificial volume force introduced by the IBM, and $\mathbf{f}_{\text{osc}}$ an oscillating force as described in section 3.

Particle equation of motion without particle-particle interaction:

$$m_p \frac{\mathrm{d}\mathbf{u}_p}{\mathrm{d}t} = \underbrace{V_p(\rho_p - \rho_f) g \mathbf{e}_g}_{\mathbf{F}_g} + \underbrace{(-3\pi D_p \mu_f \mathbf{u}_p)}_{\mathbf{F}_h} \tag{3}$$

where m_p is the particle mass, $\mathbf{u}_p$ the particle velocity vector, t time, V_p the particle volume, ρ_p and ρ_f are the densities for the particle and the fluid, respectively, g the gravitational acceleration, D_p the particle diameter, μ_f the dynamic fluid viscosity, $\mathbf{F}_g$ the weight and $\mathbf{F}_h$ the hydrodynamic force. The latter is calculated by assuming the Stokes drag, which is valid due to the application of small Reynolds numbers, as presented below.

We introduce characteristic scales for the dimensional quantities

$$\mathbf{u} = u_s \tilde{\mathbf{u}}_p = \frac{(\rho_p - \rho_f) D_p^2 g}{18\mu_f} \tilde{\mathbf{u}}_p \quad , \tag{4a}$$

$$L = D_p \tilde{L} \qquad . \tag{4b}$$

$$t = \frac{D_p}{u_s}\tilde{t} \qquad , \tag{4c}$$

$$\rho = \rho_f \tilde{\rho}_p \qquad , \tag{4d}$$

$$m_p = m_f \tilde{m}_p = \rho_f V_p \tilde{m}_p \qquad , \tag{4e}$$

$$V_p = {D_p}^3 \, \tilde{V}_p \qquad , \tag{4f}$$

$$g = \frac{u_s^2}{D_p}\tilde{g} \qquad , \tag{4g}$$

$$\mu_f = D_p u_s \rho_f \tilde{\mu}_f \qquad , \tag{4h}$$

and yield the non-dimensional Navier-stokes equations for an incompressible Newtonian fluid,

$$\frac{\partial \tilde{\mathbf{u}}}{\partial \tilde{t}} + \tilde{\nabla} \cdot (\tilde{\mathbf{u}}\tilde{\mathbf{u}}) = -\tilde{\nabla}\tilde{p} + \frac{1}{Re}\tilde{\nabla}^2 \tilde{\mathbf{u}} + \tilde{\mathbf{f}}_{IBM} + \tilde{\mathbf{f}}_{osc} \qquad , \tag{5}$$

and the non-dimensional continuity equation

$$\tilde{\nabla} \cdot \tilde{\mathbf{u}} = 0 \qquad . \tag{6}$$

Moreover, we combine (3) with the non-dimensional quantities (4) to obtain:

$$\left({D_p}^3 \rho_f \frac{u_s^2}{D_p}\right) \tilde{m}_p \frac{\mathrm{d}\tilde{\mathbf{u}}_p}{\mathrm{d}\tilde{t}} = \left({D_p}^3 \rho_f \frac{{u_s}^2}{D_p}\right) \underbrace{\tilde{V}_p(\tilde{\rho}_p - \tilde{\rho}_f)\tilde{g}\,\mathbf{e}_g}_{\tilde{\mathbf{F}}_g} + \left({D_p}^2 \rho_f {u_s}^2\right) \underbrace{\left(-3\pi \tilde{D}_p \tilde{\mu}_f \tilde{\mathbf{u}}_p\right)}_{\tilde{\mathbf{F}}_h} \qquad , \tag{7}$$

which yields

$$\tilde{m}_p \frac{\mathrm{d}\tilde{\mathbf{u}}_p}{\mathrm{d}\tilde{t}} = \left(\frac{\cancel{{D_p}^3 \rho_f \frac{{u_s}^2}{D_p}}}{\cancel{{D_p}^3 \rho_f \frac{{u_s}^2}{D_p}}}\right) \tilde{\mathbf{F}}_g + \left(\frac{\cancel{D_p^2 {u_s}^2 \rho_f}}{\cancel{{D_p}^3 \rho_f \frac{{u_s}^2}{D_p}}}\right) \tilde{\mathbf{F}}_h \qquad , \tag{8}$$

which results in

$$\tilde{m}_p \frac{\mathrm{d}\tilde{\mathbf{u}}_p}{\mathrm{d}\tilde{t}} = \tilde{\mathbf{F}}_g + \tilde{\mathbf{F}}_h \qquad . \tag{9}$$

For the parametrization of the numerical code, we need a reference density and length, as well as a reference velocity and viscosity. The reference density and length are set as $\tilde{\rho}_{\mathrm{ref}} = 1$ and $\tilde{L}_{\mathrm{ref}} = \tilde{D}_p = 1$, while the dimensionless reference velocity $\tilde{u}_{\mathrm{ref}}$ is calculated with the following equation.

$$\tilde{u}_{\mathrm{ref}} = \frac{(\tilde{\rho}_s - 1)\tilde{D}_p^2 \tilde{\mathbf{e}}_g}{18\tilde{\mu}_f} \qquad , \tag{10}$$

Table 1: Physical parameters of the dimensional and non-dimensional problem of a single sphere settling under gravity.

Parameter	dimensional			non-dimensional		
Particle diameter	D_p	$1 \cdot 10^{-5}$	m	$\tilde{D}_p$	1	-
Particle density	ρ_p	3000	kg/m^3	$\tilde{\rho}_s$	16514.8	-
Fluid density	ρ_f	1000	kg/m^3	$\tilde{\rho}_{\text{ref}}$	1	-
Dynamic viscosity	μ_f	$1 \cdot 10^{-3}$	Pa s	$\tilde{\mu}_f$	917.4	(14)
Gravity	$\mathbf{g}$	$(0, -9.81, 0)^T$	m/s^2	$\tilde{\mathbf{e}}_g$	$(0, -1, 0)^T$	-
Particle mass	m_p	$1.57 \cdot 10^{-12}$	kg	$\tilde{m}_p$	16514.8	-
Particle volume	V_p	$5.24 \cdot 10^{-16}$	m^3	$\tilde{V}_p$	1	-
Settling velocity	u_s	0.000109	m/s	$\tilde{u}_{\text{ref}}$	-1	(10)
Reynolds number	Re_r	0.00109	-	Re_s	0.00109	(12)

where $\tilde{\rho}_s = \rho_p/\rho_f$. Calculating $\tilde{\mu}_f$ and the fully transfer of the real system into the simulation system it requires a second equation. In our case, this would be the Reynolds number for the dimensional scenario

$$\text{Re}_r = \frac{\rho_f u_{\text{ref}} D_p}{\mu_f} \tag{11}$$

that has to be equal to the Reynolds number of the simulation scenario

$$\text{Re}_s = \frac{\tilde{\rho}_{\text{ref}} \tilde{u}_{\text{ref}} \tilde{D}_p}{\tilde{\mu}_f} \quad . \tag{12}$$

Using these expressions, we obtain the datasets for the real/dimensional and simulation setup as summarized in table 1.

Using (10) together with (12) yields

$$\text{Re}_s = \frac{(\tilde{\rho}_s - 1)\tilde{D}_p^2 \tilde{D}_p}{18\tilde{\mu}_f^2} \tag{13}$$

Choosing specific values for either D_p or ρ_s allows us to solve the reference dynamic viscosity

$$\tilde{\mu}_f = \sqrt{\frac{(\tilde{\rho}_s - 1)\tilde{D}_p^2 \tilde{D}_p}{18\text{Re}_s}} \tag{14}$$

This relationship makes sure that we are simulating the same particle Reynolds number for all simulations regardless of our simulation scenario. This measure also rescales our reference velocity to values different from unity as a quantity that is derived from D_p and ρ_s. Varying D_p or ρ_s, however, will change the Reynolds number Re_{PARTIES} implemented into PARTIES. This measure can be used to choose a set of nondimensional parameters that are numerically stable.

$$Re_{\text{PARTIES}} = \frac{1}{\tilde{\mu}_f} \tag{15}$$

Table 2: Physical parameters of the dimensional and non-dimensional problem of a single sphere with a non-dimensional diameter $\tilde{D}_p = 1.0$ settling under gravity.

Parameter	dimensional			non-dimensional		
Particle diameter	D_p	$1 \cdot 10^{-5}$	m	$\tilde{D}_p$	1.0	-
Particle density	ρ_p	3000	kg/m^3	$\tilde{\rho}_s$	3	-
Fluid density	ρ_f	1000	kg/m^3	$\tilde{\rho}_{\text{ref}}$	1	-
Dynamic viscosity	μ_f	$1 \cdot 10^{-3}$	Pa s	$\tilde{\mu}_f$	10.0963	(14)
Gravity	$\mathbf{g}$	$(0, -9.81, 0)^T$	m/s^2	$\tilde{\mathbf{e}_g}$	$(0, -1, 0)^T$	-
Settling velocity	u_s	0.000109	m/s	$\tilde{u}_{\text{ref}}$	-0.011005	(10)
Reynolds number	Re_r	0.00109	-	$\text{Re}_{\text{PARTIES}}$	0.099	(15)

Table 3: Physical parameters of the dimensional and non-dimensional problem of a single sphere with a non-dimensional diameter $\tilde{D}_p = 0.1$ settling under gravity.

Parameter	dimensional			non-dimensional		
Particle diameter	D_p	$1 \cdot 10^{-5}$	m	$\tilde{D}_p$	0.1	-
Particle density	ρ_p	3000	kg/m^3	$\tilde{\rho}_s$	3	-
Fluid density	ρ_f	1000	kg/m^3	$\tilde{\rho}_{\text{ref}}$	1	-
Dynamic viscosity	μ_f	$1 \cdot 10^{-3}$	Pa s	$\tilde{\mu}_f$	0.3192	(14)
Gravity	$\mathbf{g}$	$(0, -9.81, 0)^T$	m/s^2	$\tilde{\mathbf{e}_g}$	$(0, -1, 0)^T$	-
Settling velocity	u_s	0.000109	m/s	$\tilde{u}_{\text{ref}}$	-0.003480	(10)
Reynolds number	Re_r	0.00109	-	$\text{Re}_{\text{PARTIES}}$	3.132	(15)

2 Rescaling of the non-dimensionalized two-way coupled particle equation of motion

Author: Fabian (11. May 2020)

The implementation of the non-dimensional values (table 1) led to aborted simulations and therefore, a rescaling of the nondimensional values is required. By taking $\tilde{\rho}_s = 3$ and $\tilde{D}_p$ equals 1.0 and 0.1, respectively, $\tilde{u}_{\text{ref}}$, $\tilde{\mu}_f$, and Re_{PARTIES} are calculated as shown in (10), (14), and (15).The datasets for the real/dimensional and the rescaled simulation/non-dimensional set up are listed in table 2 and 3.

Based on the chosen non-dimensional particle diameters $\tilde{D}_p = 1.0$ and $\tilde{D}_p = 0.1$, the non-dimensional viscosity $\tilde{\mu}_f$, settling velocity $\tilde{u}_s$ and simulation Reynolds number Re_s have been calculated for a variety of the physical diameter D_p, as presented in table 4. These sets of values all represent feasible simulation scenarios that stably run in PARTIES. In the next sections, we will no longer consider a particles settling under gravity, but address the impact of an oscillating fluid force on the particle displacement in a zero-gravity environment.

Table 4: Numerical values of the non-dimensional problem of a single sphere settling under gravity.

D_p	$\tilde{D}_p = 1.0$			$\tilde{D}_p = 0.1$		
$[10^{-5}\ m]$	$\tilde{\mu}_f$	$\tilde{u}_s$	Re_s	$\tilde{\mu}_f$	$\tilde{u}_s$	Re_s
1.0	10.0964	0.0110	0.0990	0.3193	0.0035	3.1321
2.0	3.5696	0.0311	0.2801	0.1129	0.0098	8.8589
3.0	1.9430	0.0572	0.5147	0.0614	0.0181	16.2748
4.0	1.2620	0.0880	0.7924	0.0399	0.0278	25.0567
5.0	0.9030	0.1230	1.1074	0.0286	0.0389	35.0179
6.0	0.6870	0.1617	1.4557	0.0217	0.0511	46.0322
7.0	0.5452	0.2038	1.8343	0.0172	0.0645	46.0322
8.0	0.4462	0.2490	2.2411	0.0141	0.0787	70.8712
9.0	0.3739	0.2971	2.6742	0.0118	0.0940	84.5665
10.0	0.3193	0.3480	3.1321	0.0101	0.1101	99.0454

3 Oscillation Study

Author: Fabian (07. May 2020)

The investigation of the impact of the ISS oscillation on the particle trajectory is conducted by implementing a non-dimensional oscillation force $\tilde{f}_{\text{osc}}$, which is idealized by a sine-function, into the Navier-stokes equation (1). Thereby, two concepts will be examined, an oscillation force acting on the fluid and an oscillation force acting on the particle itself. The dimensionless function of the oscillation force (shown in (16)) is the same for both concepts, with k_1 the amplitude-constant, u_s the settling velocity, D_p the physical particle diameter, t the time, k_2 the period-constant, and $\tilde{t}$ the non-dimensional time.

$$\tilde{f}_{\text{osc}}(t) = k_1 \cdot \frac{{u_s}^2}{D_p} \cdot \sin\left(\frac{t \cdot u_s}{k_2 \cdot D_p} \cdot 2\pi\right) \cdot \mathbf{e}_x \qquad . \tag{16}$$

Note that we have computed u_s as the Stokes settling velocity for a particle settling velocity in the gravitational field of the Earth, i.e. $g = 9.81$. Hence, the peak force for $k_1 = 1$ corresponds to earth-bound conditions.

3.1 Oscillation Force Acting on Fluid

Author: Fabian (07. May 2020)

An oscillation force acting on the fluid affects the particle trajectory due to the fluid-particle interaction. The investigation of this approach is conducted for a single particle of an equivalent dimensional diameter $D_p = 10\mu m$, in zero gravity and the physical parameter as presented in table 5. The constants are chosen as $k_1 = 10{,}000$ and $k_2 = \frac{1}{545}$, respectively, to generate a steady oscillation within a short time range that we can resolve with our computational approach.

Table 5: Physical parameters for oscillation study.

Parameter	dimensional		
Particle diameter	D_p	$1 \cdot 10^{-5}$	m
Particle density	ρ_p	3000	kg/m^3
Fluid density	ρ_f	1000	kg/m^3
Dynamic viscosity	μ_f	$1 \cdot 10^{-3}$	Pa s
Gravity	$\mathbf{g}$	$(0,0,0)^T$	m/s^2
Settling velocity	u_s	0.000109	m/s
Reynolds number	Re_r	0.00109	-

First tests have shown that the oscillation requires approximately ten oscillation periods until it moves evenly around 0. Consequently, we initially fix the particle and release it after ten oscillation periods, which correlates to a non-dimensional simulation time of 0.2. Moreover, the results indicate that the courses of the fluid and particle velocities as well as the particle position in the x-direction are strongly dependent on the initial condition of the oscillation. Starting a sine-function with zero forcing as the initial condition induces a bulk flow in positive x-direction, which is unphysical given the closed containers used on the ISS. Therefore, we implemented a phase shift p_s into (16), as the following

$$\tilde{f}_{\mathrm{osc}}(t) = k_1 \cdot \frac{{u_s}^2}{D_p} \cdot \sin\left(\frac{t \cdot u_s}{k_2 \cdot D_p} \cdot 2\pi \; + \; p_s\right) \cdot \mathbf{e}_x \qquad , \tag{17}$$

and investigated the behaviour of the above mentioned courses regarding the modified initial forcing. The figures 1 and 2 illustrate the oscillating fluid and particle velocities due to the oscillation force without a phase shift ($p_s = 0$) and with a phase shift $p_s = \frac{\pi}{2}$, respectively. It is evident that the initial as well as the stabilized fluid velocity is higher for $p_s = 0$ than for $p_s = \frac{\pi}{2}$. Consequently, the average fluid velocity for $p_s = 0$ is seven times higher with $\overline{u}_{\mathrm{f}} = 0.007$ than for $p_s = \frac{\pi}{2}$ ($\overline{u}_{\mathrm{f}} = 0.001$), where overbar indicates time average over the entire simulation time.

The comparison of the particle position courses of $p_s = 0$ and $p_s = \frac{\pi}{2}$, shown in figure 3, illustrate that the movement of the particle is also highly dependent on the initial oscillation condition.

Further simulations have been conducted to examine the effect of the particle release on its velocity and position course. Therefore, four different times of particle releases have been chosen, the minimum, the maximum , and the zero crossing of the fluid velocity. Whereas the zero crossing has been distinguished into the upwards trending velocity (maximal acceleration) and the downwards trending velocity (minimal acceleration). Figure 4 shows the velocity courses for these four particle release times. The graph highlights that the initial fluid impact is decisive for the behavior of the particle velocity. The release times with a subsequent upwards trending velocity (minimum and zero-

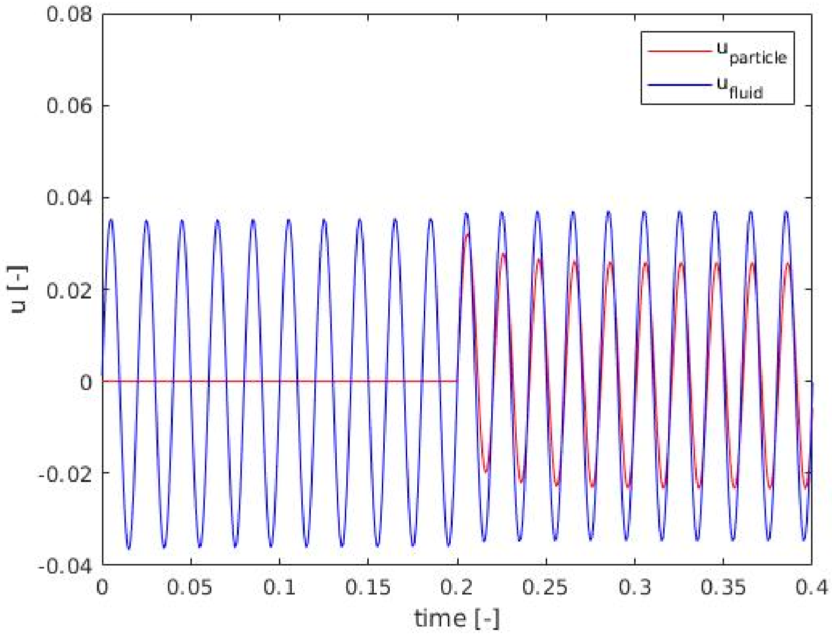

Figure 1: Fluid and particle velocity courses caused by an oscillation force without a phase shift ($p_s = 0$).

upwards) result in higher and predominantly positive averaged velocities, while the release times with a subsequent downwards trending velocity (maximum and zero-downwards) result in lower and predominantly negative averaged velocities.

On this basis, the particle position courses in the x-direction, illustrated in figure 5, also vary. The minimum and zero-upwards release times cause a particle movement in the positive x direction, while the maximum and zero-downwards release times cause a particle movement in the negative x direction.

In addition, we compared the simulations of the two-way coupled model (PARTIES) and a one-way coupled model that was developed and implemented in a Matlab-script. By using a repeating oscillation for the one-way coupled model, which is based on the calculated bulk velocity of the PARTIES simulation, we generate similar fluid conditions. The comparison of the velocities is illustrated in figure 6. One of the main difference between the one- and two-way coupled model is visible after the release of the particle. While the oscillation of the one-way coupled fluid velocity remains constant, the two-way coupled fluid velocity moves upward to higher velocities. Moreover, the oscillation range of the two-way coupled particle velocity is larger than for the one-way coupled particle velocity and shows a phase shift later in time. However, it is important to mention that the equations of the Matlab-model are discretized with

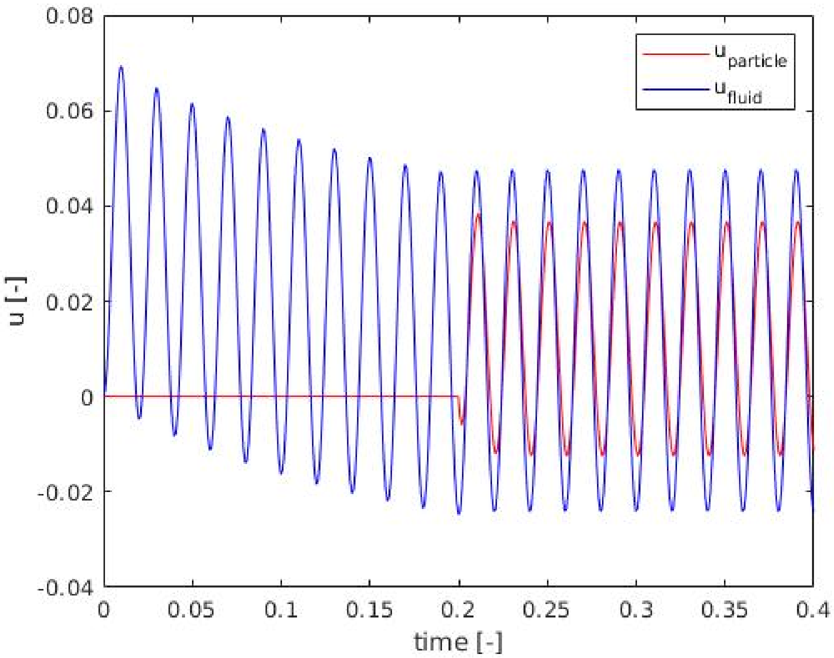

Figure 2: Fluid and particle velocity courses caused by an oscillation force with a phase shift $p_s = \frac{\pi}{2}$.

an Euler-forward scheme of first order accuracy, while in PARTIES they are discretized with third order accuracy (Runge-Kutta 3 step).

Considering the particle trajectory in the x-direction of the models (figure 7) highlights the divergent behaviours of the particles. While the particle of the two-way coupled model constantly moves in positive x-direction, the one-way coupled particle swings back after an initial movement into the positive x-direction.

We assume that the one-way coupled model might not be sufficient for reproducing the particle-fluid interaction with an implemented oscillation. Nevertheless, the presented results of the simulations in PARTIES give reason to suppose that the periodic boundaries affect the particle and fluid results. Especially the comparison of the particle courses in x-direction of leads to the assumption that the simulation in PARTIES generates a background flow and, therefore, results in a constant movement of the particle.

To conclude, our studies show that the approach of an oscillating fluid-volume force produces results that are highly dependent on the initial condition, both for the fluid and the particle motion, and are also highly affected by the boundary conditions. We therefore have concerns that this is not a very robust

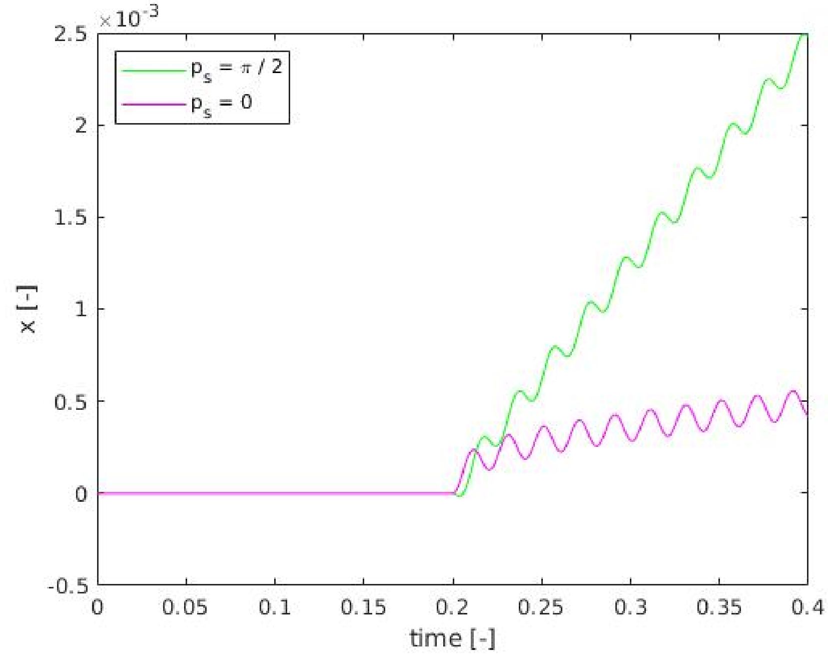

Figure 3: Comparison of the particle position courses in the x-direction caused by an oscillation force.

approach to generate physical results.

3.2 Oscillation Force Acting on the Particle

Author: Fabian (18. May 2020)

In this section, the application of an oscillation force acting on the particle is presented. Therefore, we take the non-dimensional particle equation of motion (9) and add the oscillation force acting on the particle. Due to zero gravity, we obtain

$$\tilde{m}_p \frac{\mathrm{d}\tilde{\mathbf{u}}_p}{\mathrm{d}\tilde{t}} = \cancel{\tilde{\mathbf{F}}_g} + \tilde{\mathbf{F}}_h + \tilde{\mathbf{F}}_{\mathrm{osc}} \qquad , \tag{18}$$

where

$$\tilde{\mathbf{F}}_{\mathrm{osc}}(t) = k_1 \cdot \frac{{u_s}^2}{D_p} \cdot \sin\left(\frac{t \cdot u_s}{k_2 \cdot D_p} \cdot 2\pi \ + \ p_s\right) \cdot \mathbf{e}_x \qquad . \tag{19}$$

Similar to the previous approach of the fluid oscillation force, we take the constants as $k_1 = 10,000$ and $k_2 = \frac{1}{545}$ to generate a steady oscillation within

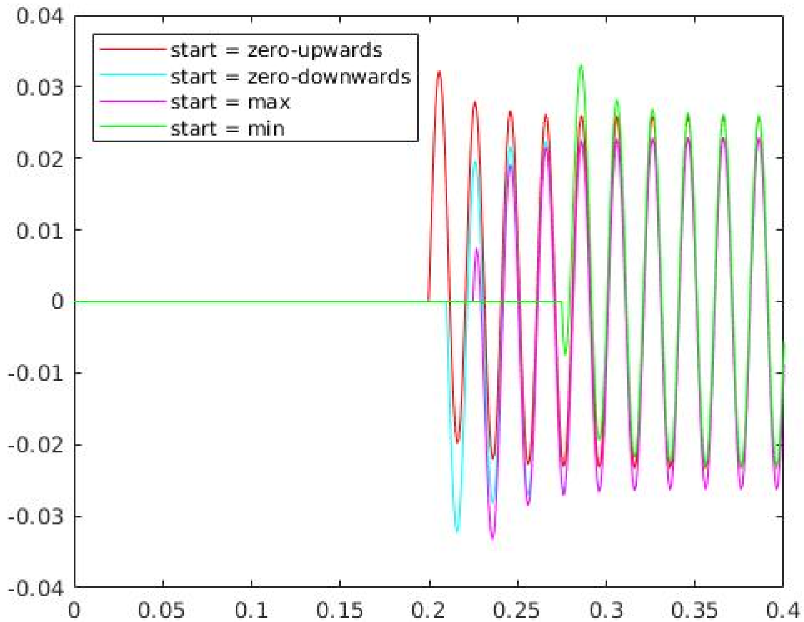

Figure 4: Velocity over time for four different particle release times with oscillating force on the fluid (5).

time range short enough that we can resolve it. In the first tests we investigated the effect of a phase shift on the velocities and particle courses. As visible in figure 8 and 9, the velocities need approximately five oscillation periods ($time = 0.1[-]$) to even out after initial deviations. To illustrate the constant course of the velocities after the initial period, we computed the averaged velocities, as presented in table 6. Based on these averaged oscillation velocities, it seems to be that they are independent of the phase shift. Whereas the initial movement of the particle courses, presented in figure 10, might be strongly dependent on the initial forcing condition.

Consequently, we examined several simulations with initial fixed particles and varying releasing times. Similar to the tests of the fluid oscillations, we distinguished the releasing times into minimum, maximum, zero upwards (maximal acceleration), and zero downwards (minimal acceleration). The results show the same outcome as for the phase shift analysis with consistent particle and fluid velocities after initial divergences (figure 11 and 12), and divergent particle courses, as shown in figure 13. The presented courses in the x-direction highlight that the particle of each scenario experiences an initial shift in the respective direction and oscillates after approximately five oscillation periods at the same location. Therefore, it is important to emphasize that the beginning

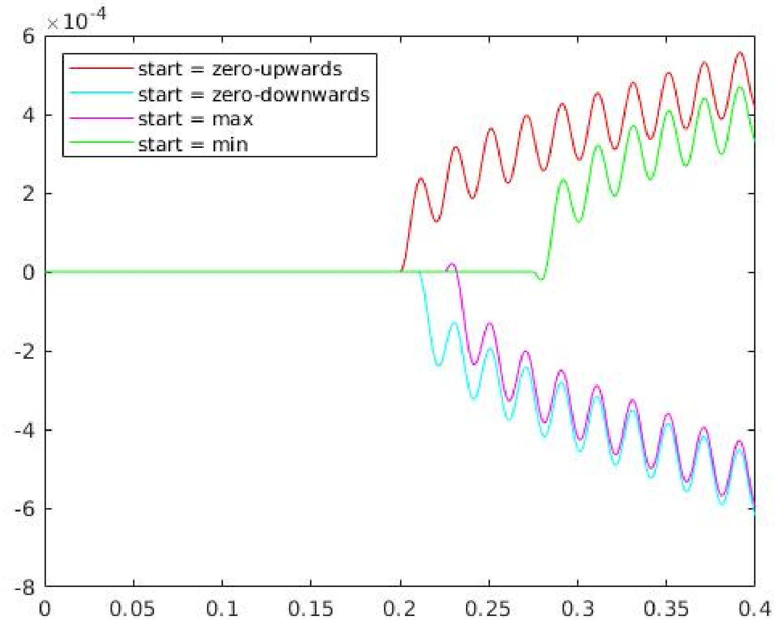

Figure 5: Position courses in the x- direction for four different particle release times with oscillating force on the fluid (5).

Table 6: Averaged oscillation velocities.

phase shift p_s	$\overline{u}_{\mathrm{p}}[-]$	$\overline{u}_{\mathrm{f}}[-]$
0	$1.11 \cdot 10^{-5}$	$-3.21 \cdot 10^{-8}$
0.5π	$1.46 \cdot 10^{-6}$	$-2.88 \cdot 10^{-8}$
π	$-8.81 \cdot 10^{-6}$	$-1.58 \cdot 10^{-8}$

of the particle movement is affected by the initial conditions (phase shift and release time), but they reach a consistent movement after the above mentioned initial period.

Furthermore, we compared our two-way coupled simulations (PARTIES) with the accompanying one-way coupled model (Matlab). The results, presented in figure 14 and 15, indicate that the velocities and the particle courses are basically similar. The magnitude of the one-coupled particle velocity is one and a half times higher than the of the two-way coupled velocity, which could be due to the fact that the particle-fluid interaction is not considered.

Summing up the studies of the implementation of an oscillation force acting on the particle, we observe that the initial conditions only affect the particle within a initial period of approximately five oscillation periods. After this pe-

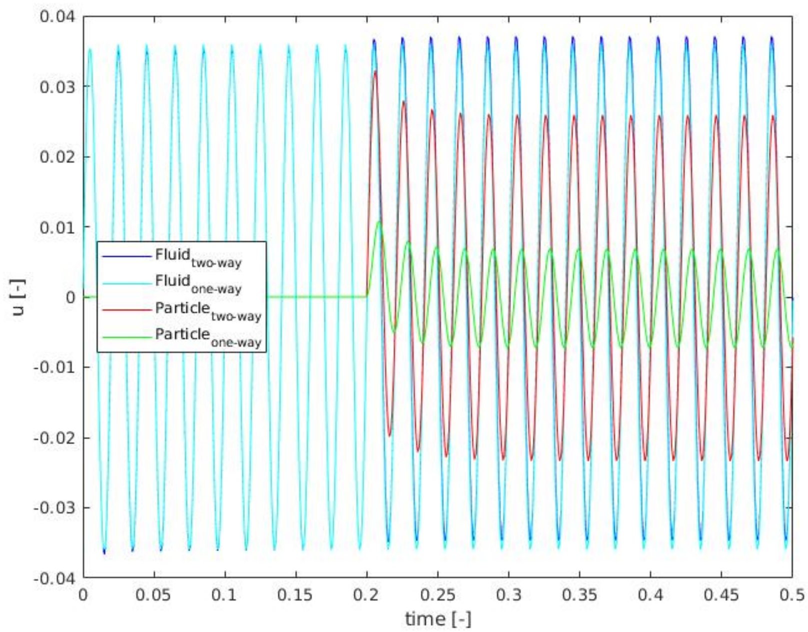

Figure 6: Comparison of the one- and two-way coupled fluid and particle velocities with oscillating forcing of the fluid (5).

riod, the particle velocity and movement is independent of these conditions and similar for all investigated scenarios.

4 Key Findings so far

The key findings of the present study and basis for further discussions are:

Fluid oscillation force:

- strong dependency of the simulation results on the initial conditions
- effect of the periodic boundary conditions on the fluid and particle motion
- a possible unintentionally generated background flow
- strong deviation between using Stokes-drag (Matlab-script) and hydrodynamic forces computed by the IBM.

Particle oscillation force:

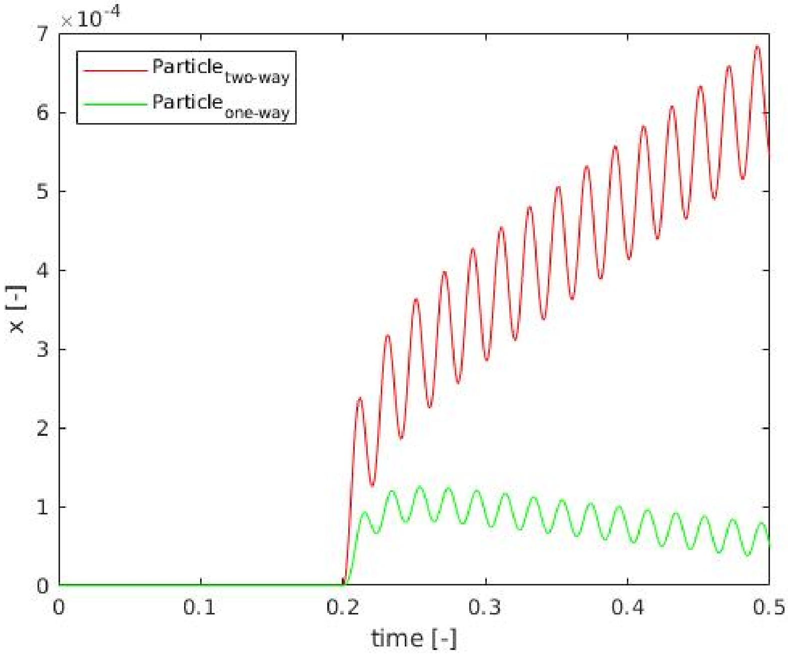

Figure 7: Comparison of the one- and two-way coupled particle courses in the x-direction with oscillating forcing of the fluid (5)..

- negligible dependency of the simulation results on the initial conditions
- similar trend of the particle velocities and courses after the initial period of five oscillation periods
- similar trends of the one- and two-way coupled simulations.

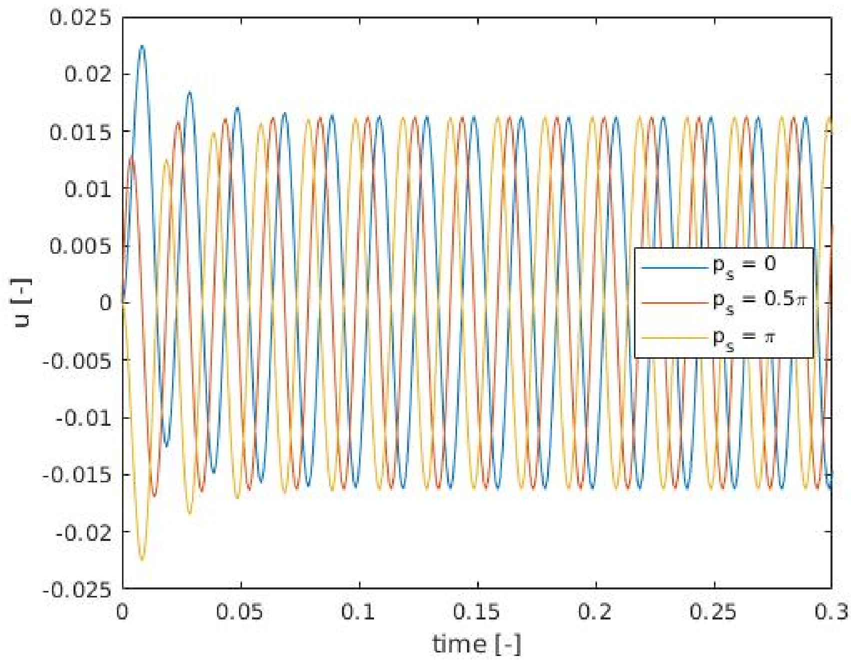

Figure 8: Particle velocities depending on phase shift p_s.

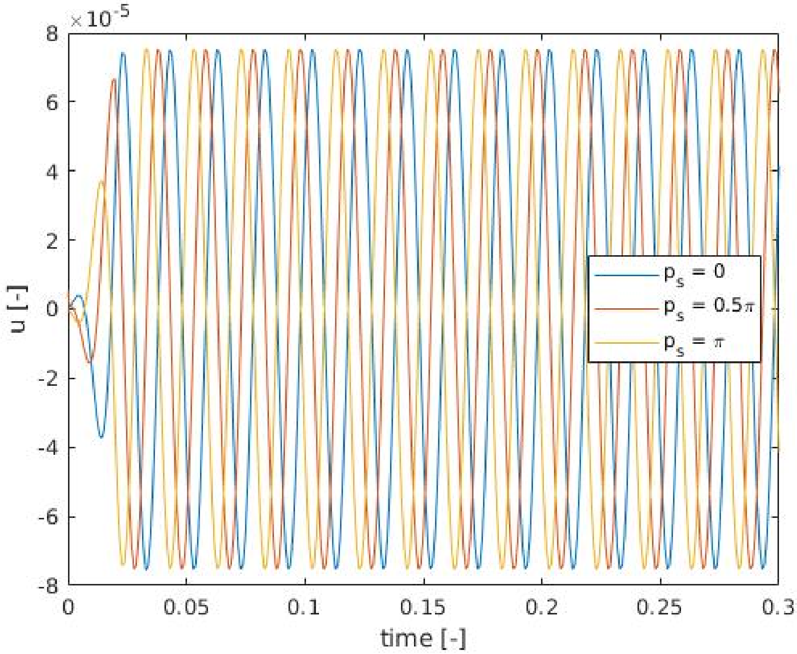

Figure 9: Fluid velocities depending on phase shift p_s.

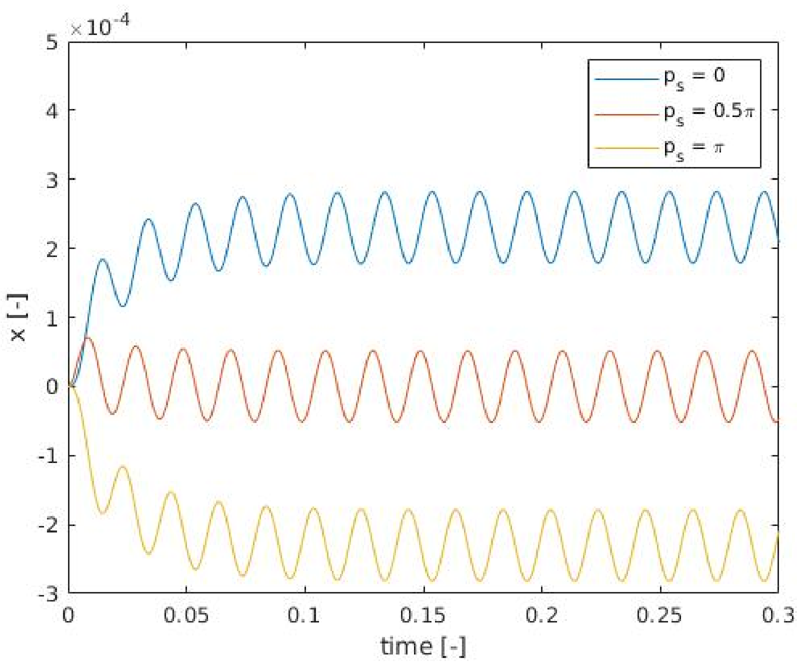

Figure 10: Particle course in x-direction depending on phase shift p_s.

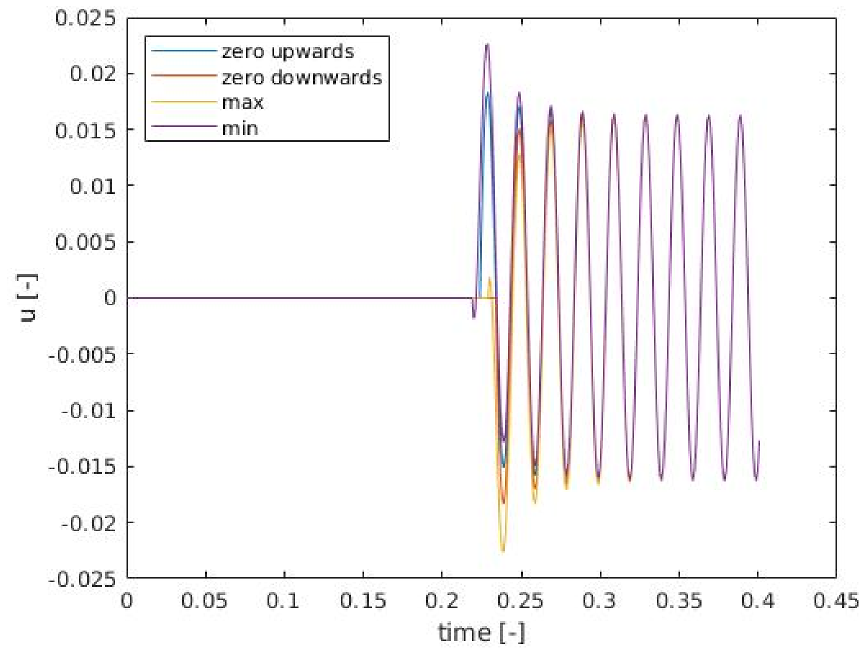

Figure 11: Particle velocities depending on release time.

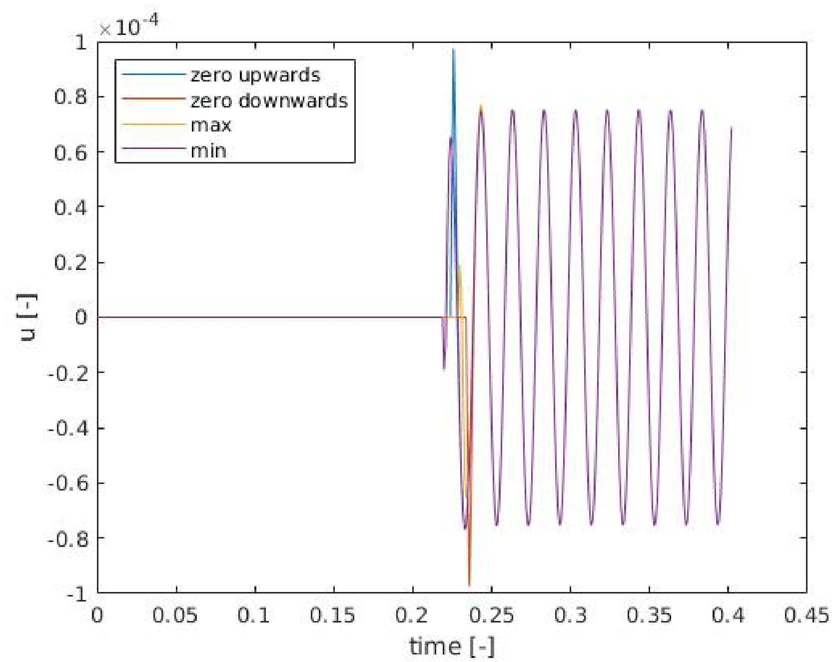

Figure 12: Fluid velocities depending on release time.

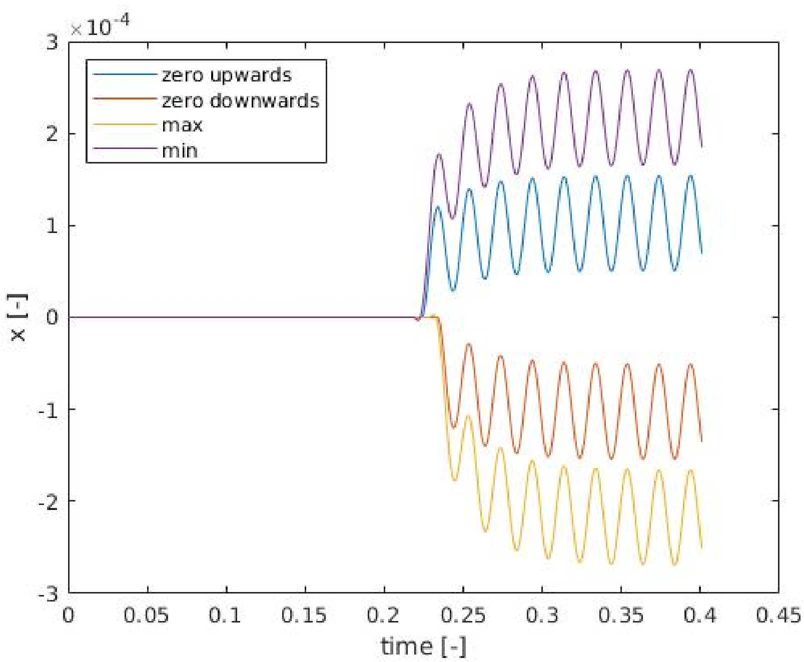

Figure 13: Particle course depending on release time.

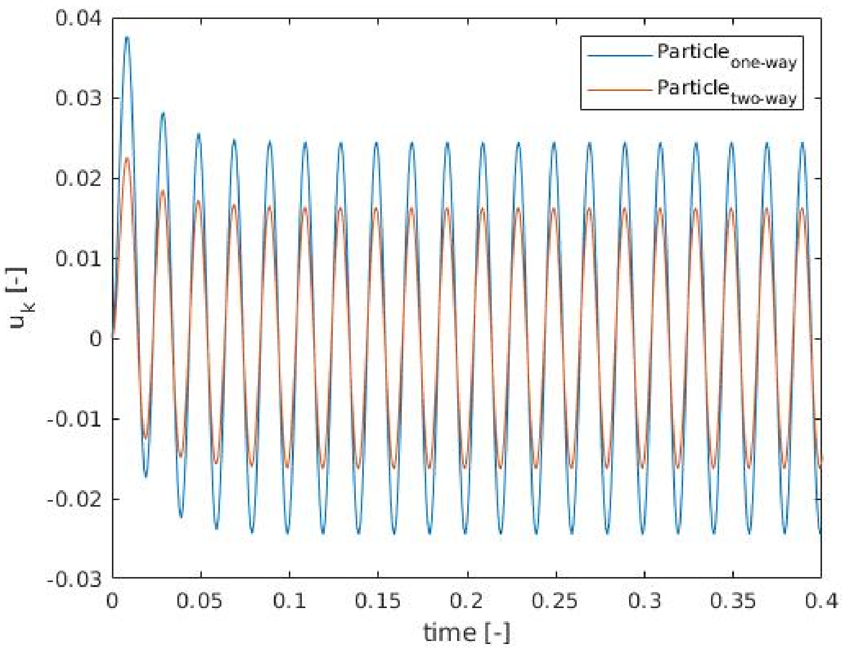

Figure 14: Comparison of the one- and two-way coupled particle velocities.

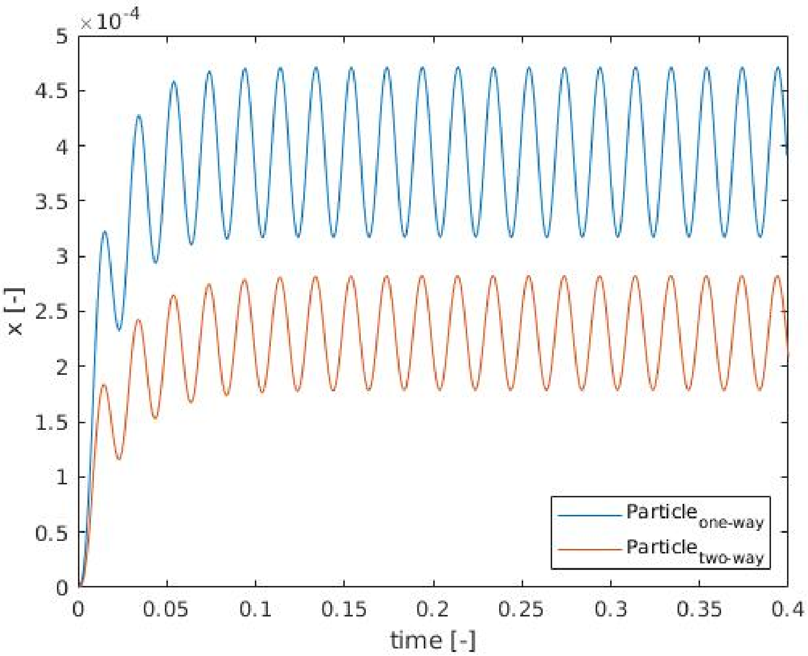

Figure 15: Comparison of the one- and two-way coupled particle courses in the x-direction.